

If $n > 1$, the function represents the increasing returns to scale. In other words, if all the inputs are increased by one percent, the resultant increase in output will be more than one percent.

If $n < 1$, the function connotes the diminishing returns to scale. It means, if all inputs are increased by one percent, the increase in output will be less than one percent.

(II) LINEAR HOMOGENEOUS PRODUCTION FUNCTION

In any homogeneous production function, if the degree of homogeneity is equal to one, the function is said to be linear homogeneous production function. Here, it is to be remembered that word "linear" should not be confused with the linear relation between the variables of the function. A function can be linear homogeneous and yet it may not be linear in relation as

$$u = f(x, y) = \sqrt{x^2 + 3xy + y^2}$$

is a linear homogeneous but not linear in relation. Thus, the word linear here should always be interpreted only as a synonymus of the first degree.

Linear homogeneous production function is frequently used in the theory of production. Suppose, $Q = f(L, C)$ be a production function. It will be the linear production function, if $(tL, tC) = tQ$, for all values off.

COBB-DOUGLAS PRODUCTION FUNCTION

Cobb-Douglas production function is known after the names of American Economists C. W. Cobb and P.H. Douglas. It is based on the assumption that output is a function of labour and capital. In its original version, Cobb-Douglas production function is linear homogeneous function of degree one. More precisely, this form of function is based on the empirical study of the American manufacturing industry as a whole for the period 1899-1922. It assumes that American manufacturing industry was operating under constant returns to scale. The general version of Cobb-Douglas production function is :

$$Q = AL^\alpha C^\beta$$

Here, Q is the quantity of output, L stands for quantity of labour and C for capital. A , α and β are the parameters.

Roughly speaking, Cobb-Douglas production function states that about three fourth of the increase in manufacturing output is due to labour input and the remaining one-fourth is due to capital input.

No doubt, Cobb-Douglas production function in its original sense is applied to industrial sector, but later on it was applied in agricultural sector too. The earliest application of this type of function appears to be by Kaniya of the university of Tokyo in 1941. In this very attempt one restraint i.e. $\alpha + \beta = 1$ was relaxed and the values of α & β were determined independently of each other. Since then, Cobb-Douglas production function has been applied in agricultural sector on an extensive scale. Now, many inputs are being used as independent variables in such a production function. Now a days, the production function which is in use can be described in the following form :

$$y = ax_1^{b_1} x_2^{b_2}$$

Here $(b_1 + b_2 + \dots)$ may or may not be equal to one. It all depends on the value of b_1 , b_2 , etc. as determined after the function has been fitted.

FEATURES OF COBB-DOUGLAS PRODUCTION FUNCTION

The salient features of the Cobb-Douglas production function are as follows :

1. Cobb-Douglas production function is homogeneous of degree $\alpha + \beta$. In the special case if $\alpha + \beta = 1$, the function becomes linear homogeneous function.

$$Q = AL^\alpha C^\beta = f(L, C)$$

$$f(tL, tC) = A (tL)^\alpha (tC)^\beta = A t^\alpha L^\alpha t^\beta C^\beta = t^{\alpha+\beta} AL^\alpha C^\beta = t^{\alpha+\beta} Q.$$

In this way, Cobb-Douglas production function is homogeneous of degree $\alpha + \beta$. In case, if $\alpha + \beta = 1$, as stated earlier, the function will be linear homogeneous which in turn will depict the constant returns to scale.

when

$$\alpha + \beta = 1$$

$$Q = AL^\alpha C^{1-\alpha} \text{ or } Q = AL^\beta C^{1-\beta}$$

2. In Cobb-Douglas production function, the marginal product of labour and capital are the first order partial derivatives of output with respect to labour and capital. Symbolically,

$$\begin{aligned} \text{Marginal Product of Labour (MP}_L\text{)} &= \frac{\partial Q}{\partial L} \\ &= \frac{\partial}{\partial L} (AL^\alpha C^\beta) = A \frac{\partial}{\partial L} L^\alpha C^\beta = \alpha AL^{\alpha-1} C^\beta \\ &= \frac{\alpha AL^\alpha C^\beta}{L} \text{ i.e. } \alpha \left(\frac{Q}{L} \right) \end{aligned}$$

$$\begin{aligned} \text{Marginal Product of Capital (MP}_C\text{)} &= \frac{\partial Q}{\partial C} \\ &= \frac{\partial}{\partial C} (AL^\alpha C^\beta) = AL^\alpha \frac{\partial}{\partial C} C^\beta = AL^\alpha \beta C^{\beta-1} \\ &= \frac{AL^\alpha \beta C^\beta}{C} \text{ i.e. } \beta \left(\frac{Q}{C} \right) \end{aligned}$$

3. In Cobb-Douglas production, elasticity of substitution is equal to one.

$$\text{Elasticity of Substitution} = \frac{\text{Proportionate change in factor - quantity ratio}}{\text{Proportionate change in factor - price ratio}}$$

4. In Cobb-Douglas production function isoquants have negative slope.

$$Q = AL^\alpha C^\beta = f(L, C)$$

Total Differential

$$dQ = \frac{\partial Q}{\partial L} dL + \frac{\partial Q}{\partial C} dC$$

Differentiate both sides partially w.r.t.L.

$$\frac{\partial Q}{\partial L} = A\alpha L^{\alpha-1} C^\beta$$

Now differentiate both sides partially w.r.t. C.

$$\frac{\partial Q}{\partial C} = A\alpha L^\alpha \beta C^{\beta-1} = A\beta L^\alpha C^{\beta-1}$$

Since output remains unchanged on a particular isoquant, the total change in output $dQ=0$.

$$\therefore dQ = (A\alpha L^{\alpha-1} C^{\beta}) dL + (A\beta L^{\alpha} C^{\beta-1}) dC$$

$$\text{i.e. } \frac{dL}{dC} = - \frac{\beta L}{\alpha C}$$

Thus isoquant possesses negative slope.

Again

$$\begin{aligned} \frac{d^2L}{dC^2} &= \frac{d}{dC} \left(-\frac{\beta L}{\alpha C} \right) = -\frac{\beta}{\alpha} \frac{d}{dC} \left(\frac{L}{C} \right) \\ &= -\frac{\beta}{\alpha} \cdot \frac{1}{C^2} \left(C \frac{dL}{dC} - L \right) = -\frac{\beta}{\alpha} \cdot \frac{1}{C} \frac{dL}{dC} + \frac{\beta}{\alpha} \frac{L}{C^2} \end{aligned}$$

5. Cobb-Douglas production function represents the share of labour and capital in output.

$$Q = AL^{\alpha} C^{\beta}$$

$$\text{Marginal Product of Labour (MP}_L\text{)} = \frac{\partial Q}{\partial L}$$

Partially differentiate Q w.r.t. L. and multiplying by L we get:

$$\begin{aligned} MP_L &= \frac{dQ}{dL} L \text{ i.e. } A \alpha L^{\alpha-1} C^{\beta} \\ &= A \alpha L^{\alpha} C^{\beta} \end{aligned}$$

From this we can obtain the labour share in output.

$$MP_L = \frac{\text{Labour share}}{\text{Total income}} = \frac{A\alpha L^{\alpha} C^{\beta}}{AL^{\alpha} C^{\beta}} = \alpha$$

$$\text{Similarly, } MP_C = \frac{\partial Q}{\partial C} \cdot C$$

Now, partially differentiate Q w.r.t. C.

$$\begin{aligned} \frac{\partial Q}{\partial C} &= AL^{\alpha} \beta C^{\beta-1} \\ MP_C &= \frac{dQ}{dC} C \text{ i.e. } AL^{\alpha} \beta C^{\beta-1} C = AL^{\alpha} \beta C^{\beta} \end{aligned}$$

From this we can get the share of capital in output.

$$MP_C = \frac{\text{Capital Share}}{\text{Total Income}} = \frac{AL^{\alpha} \beta C^{\beta}}{AL^{\alpha} C^{\beta}} = \beta$$

6. In Cobb-Douglas production function α and β represent the elasticities of output with respect to labour and capital i.e.

$$\text{Elasticity of output w.r.t. Labour} = \frac{\text{Proportionate change in output}}{\text{Proportionate change in labour}}$$

$$\text{i.e. } \frac{\frac{\partial Q}{\partial L} \cdot L}{Q} = \frac{\partial Q}{Q} \times \frac{L}{\partial L} = \frac{L}{Q} = \frac{\partial Q}{\partial L} \quad \left[\because \frac{\partial Q}{\partial L} = MP_L \right]$$

Therefore

$$= \frac{L}{Q} MP_L = \frac{L}{Q} \alpha \left(\frac{Q}{L} \right) = \alpha \frac{L}{Q} \frac{Q}{L} = \alpha$$

In the same way ;

$$\text{Elasticity of output w.r.t. capital} = \frac{\text{Proportionate change in output}}{\text{Proportionate change in capital}}$$

7. Cobb-Douglas production function satisfies the Euler's Theorem which states

$$MP_L L + MP_C C = nQ$$

REASONS FOR THE POPULARITY OF COBB-DOUGLAS PRODUCTION FUNCTION

1. Cobb-Douglas production function assumes that all inputs are complementary at the minimum level.
2. By giving this form of production function one can determine whether the returns to scale are constant, increasing or decreasing.
3. If two inputs are assumed for fitting the production function and $a + B = 1$, the values of these two coefficients depict the relative share of each factor in the total output.
4. This type of production function facilitates to know the elasticities of production of various factors.

Returns to Scale : The return to a factor can be worked out on the basis of its regression coefficient. For example if $R^2 = 0.90$, it indicates that 90% of the variations in y are explained by the independent variables taken in the function. Further the estimated function gives the results.

$$y = aL^5C^3 \quad (R^2 = 0.90)$$

The return to labour (L) will be

$$= \frac{0.5}{0.8} \times 0.90 \bar{y}$$

The return to scale on the other hand refers to the sum of contribution made by all the factors taken in the function or sum of elasticities (Σb_i)

If $\Sigma b_i = 1$ it shows constant return to scale, (figure 4.11).

If $\Sigma b_i < 1$ it shows diminishing returns to scale, (figure 4.12)

If $\Sigma b_i > 1$ it shows increasing returns to scale, (figure 4.13)

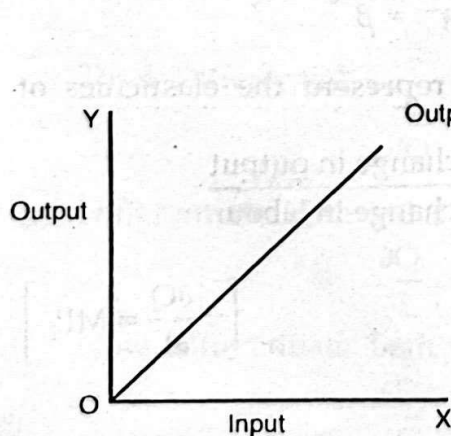


Fig. 4.11

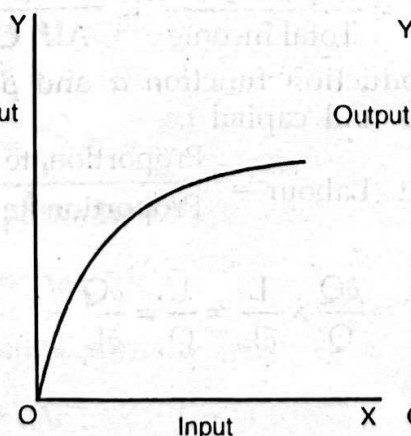


Fig. 4.12

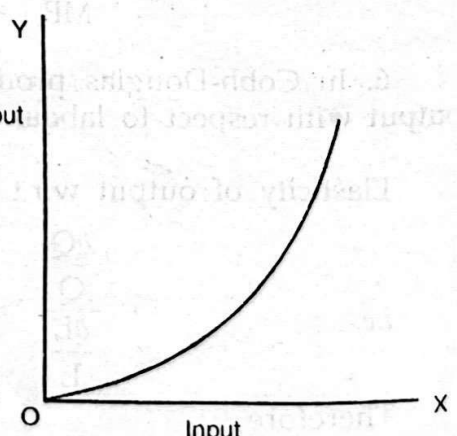


Fig. 4.13